PROBLEM SET 1 DUE 3.15

All rings are commutative with unit unless expressly stated otherwise.

1. Let $A = \mathbb{Z}[x, y]$ and let $\mathfrak{a} = (x + y)$, $\mathfrak{b} = (x)$, $\mathfrak{c} = (y)$ be ideals of A. Show that

$$\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) \supsetneq \mathfrak{a} \cap \mathfrak{b} + \mathfrak{a} \cap \mathfrak{c}.$$

2. Find a counterexample to the statement "$\mathfrak{ab} = \mathfrak{a} \cap \mathfrak{b}$ implies $\mathfrak{a} + \mathfrak{b} = (1)$".

3. Let $\mathfrak{a}, \mathfrak{b}$ be ideals of a ring A.

(1) Show that $(\mathfrak{a} : \mathfrak{b})/\mathfrak{a} = \mathrm{Ann}_{A/\mathfrak{a}}\Big((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}\Big)$.
(2) Show that $(\mathfrak{a} : \mathfrak{b})\mathfrak{b} \subseteq \mathfrak{a}$ and equality holds if $\mathfrak{a} \subseteq \mathfrak{b}$ and $\mathfrak{b}$ is principal.
(3) Calculate $(m\mathbb{Z} : n\mathbb{Z})$ and $(m\mathbb{Z} : n\mathbb{Z})(n\mathbb{Z})$.

4. Show that ab is a zero-divisor if and only if a is a zero-divisor or b is a zero-divisor.

5. Show that if u is invertible and x is nilpotent, then $u + x$ is invertible.

6. Let $A = k[x, y]/(xy)$ where k is a field. Let $f(x, y) \in k[x, y]$. Prove the following statements:

(1) $\bar{f}$ is a unit in A if and only if $\bar{f} = \bar{c}$ for some $c \in k^\times$.
(2) $\bar{f}$ is a zero-divisor in A if and only if $x | f(x, y)$ or $y | f(x, y)$.
(3) $\bar{f}$ is a nilpotent element in A if and only if $\bar{f} = 0$.

7. An ideal $\mathfrak{a}$ is **nilpotent** if $\exists n \geqslant 1$ such that $\mathfrak{a}^n = 0$. If $\mathfrak{a}$ is nilpotent, then $\mathfrak{a}$ consists of nilpotent elements. Prove the following statements:

(1) If $\mathfrak{a}$ is generated by nilpotent elements, then $\mathfrak{a}$ consists of nilpotent elements.
(2) If $\mathfrak{a}$ is generated by finitely many nilpotent elements, then $\mathfrak{a}$ is nilpotent. In particular, a finitely generated ideal consisting of nilpotent elements is nilpotent.
(3) Find an example of a non-nilpotent ideal $\mathfrak{a}$ consisting of nilpotent elements.
(4) Find an example of a non-nilpotent ideal $\mathfrak{a}$ for which $\exists n \geqslant 1$ such that $x^n = 0$ for all $x \in \mathfrak{a}$.

8. Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x with coefficients in A. Let $f(x) = a_0 + a_1 x + \ldots + a_n x^n \in A[x]$. Show that

(1) $f(x)$ is a unit in $A[x]$ if and only if a_0 is a unit in A and $a_1, \ldots, a_n$ are nilpotent.
(2) $f(x)$ is nilpotent if and only if $a_0, a_1, \ldots, a_n$ are nilpotent.
(3) $f(x)$ is a zero-divisor if and only if there exists $a \neq 0$ in A such that $af(x) = 0$.
(4) $f(x)$ is said to be **primitive** if $(a_0, a_1, \ldots, a_n) = (1)$. Show that if $f(x), g(x) \in A[x]$, then $f(x)g(x)$ is primitive $\Leftrightarrow$ $f(x)$ and $g(x)$ are primitive.

See Atiyah-MacDonald (A-M) Exercise 1.2 for hints.

9. Let $\{\mathfrak{p}_i\}_{i \in I}$ be a family of prime ideals in A such that for any $i, j \in I$, one has $\mathfrak{p}_i \subseteq \mathfrak{p}_j$ or $\mathfrak{p}_j \subseteq \mathfrak{p}_i$. Show that $\bigcup_{i \in I} \mathfrak{p}_i$ and $\bigcap_{i \in I} \mathfrak{p}_i$ are both prime ideals of A.

10. Let A, B be rings.

(1) Show that every ideal of $A \times B$ has the form

$$\mathfrak{a} \times \mathfrak{b} = \{(x, y) : x \in \mathfrak{a}, y \in \mathfrak{b}\}$$

for some ideals $\mathfrak{a} \subseteq A$ and $\mathfrak{b} \subseteq B$.

(2) Show that every prime ideal of $A \times B$ has the form

$$\mathfrak{p} \times B \quad \text{or} \quad A \times \mathfrak{q}$$

for some prime ideals $\mathfrak{p} \subseteq A$ and $\mathfrak{q} \subseteq B$.

11. Show that $\mathfrak{a}$ is a radical ideal if and only if $\mathfrak{a}$ is an intersection of prime ideals.

12. Let $\{\mathfrak{a}_i\}_{i \in I}$ be a family of ideals in A. Is it true that

$$\sqrt{\bigcap_{i \in I} \mathfrak{a}_i} = \bigcap_{i \in I} \sqrt{\mathfrak{a}_i} \ ?$$

Prove it if the equality holds, otherwise give a counterexample.

13. Let A be a ring, x an indeterminate. Use Problem **8** to show that

$$\operatorname{nil}(A[x]) = \operatorname{rad}(A[x]) = (\operatorname{nil} A)[x] \ [1].$$

14. Let A be a ring. A subset $S \subseteq A$ is **multiplicative** if $1 \in S$ and $a, b \in S$ implies $ab \in S$. Show that if S is multiplicative and $\mathfrak{a}$ is an ideal of A such that $\mathfrak{a} \cap S = \varnothing$, then there exists a prime ideal $\mathfrak{p}$ such that $\mathfrak{a} \subseteq \mathfrak{p}$ and $\mathfrak{p} \cap S = \varnothing$.

15. Let A be a ring. A multiplicative subset $S \subseteq A$ is **saturated** if

$$xy \in S \quad \Rightarrow \quad x \in S \text{ and } y \in S.$$

Problem **4** shows that the set of non-zero-divisors in A is a saturated multiplicative subset. Use Problem **14** to show that the complement of a saturated multiplicative subset is a union of prime ideals. In particular, the set of zero-divisors in A is a union of prime ideals.

16. Let A be a ring and let $\mathfrak{a}$ be an ideal of A. A **minimal prime ideal** of A is a prime ideal $\mathfrak{p}$ for which there exists no prime ideal $\mathfrak{q}$ with $\mathfrak{q} \subsetneq \mathfrak{p}$. We say that a prime ideal $\mathfrak{p}$ is **over** $\mathfrak{a}$ if $\mathfrak{a} \subseteq \mathfrak{p}$. A **minimal prime ideal over** $\mathfrak{a}$ is a prime ideal $\mathfrak{p}$ over $\mathfrak{a}$ for which there exists no prime ideal $\mathfrak{q}$ with $\mathfrak{a} \subseteq \mathfrak{q} \subsetneq \mathfrak{p}$.

(1) Use Problem **9** to show that every prime ideal of A contains a minimal prime ideal.

(2) Show that every prime ideal over $\mathfrak{a}$ contains a minimal prime ideal over $\mathfrak{a}$, hence

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \text{ minimal prime over } \mathfrak{a}} \mathfrak{p}.$$

17. An element e is an **idempotent** if $e^2 = e$. Show that a local ring contains no idempotents other than 0 and 1. [2]

18. Let A be a ring and let $A[[x]]$ be the ring of formal power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ with coefficients in A. Prove the following statements:

(1) $f(x)$ is a unit in $A[[x]]$ if and only if a_0 is a unit in A.

---

[1] Given an ideal $\mathfrak{a}$ of A, $\mathfrak{a}[x]$ is the ideal of $A[x]$ consisting of polynomials whose coefficients all belong to $\mathfrak{a}$.

[2] An idempotent allows us to decompose a ring into the direct product of two rings. Explicitly, if e is an idempotent of R, then $R \cong eR \times (1-e)R$ as rings where eR is a ring with identity e and $(1-e)R$ is a ring with identity $1-e$. Conversely, if $R = A \times B$, then $(1,0)$ is an idempotent of R, so is $(0,1)$. Therefore, this exercise is equivalent to the assertion that a local ring is not the direct product of two nonzero rings.

(2) If $f(x)$ is nilpotent, then a_n is nilpotent for all $n \geqslant 0$.
(3) The converse of (2) is not true. For example, consider the ring

$$A = \mathbb{F}_2[t, t^{\frac{1}{2}}, t^{\frac{1}{4}}, \ldots]/(t)$$

and the formal power series

$$f(x) = \overline{t^{\frac{1}{2}}}x + \overline{t^{\frac{1}{4}}}x^2 + \ldots + \overline{t^{\frac{1}{2^n}}}x^n + \ldots.$$

in $A[[x]]$. Show that $f(x)$ is *not* nilpotent.

Hint: in a ring of characteristic 2, one has $(x+y)^2 = x^2 + y^2$.

(4) $f(x)$ belongs to the Jacobson radical of $A[[x]]$ if and only if a_0 belongs to the Jacobson radical of A. That is, $\operatorname{rad}(A[[x]]) = (\operatorname{rad} A, x)$.
(5) The contraction $\mathfrak{n}^c$ of a maximal ideal $\mathfrak{n}$ of $A[[x]]$ is a maximal ideal of A, and $\mathfrak{n}$ is generated by $\mathfrak{n}^c$ and x.

Hint: first show that $x \in \mathfrak{n}$.

(6) If $(A, \mathfrak{m})$ is a local ring, then $A[[x]]$ is a local ring with maximal ideal $(\mathfrak{m}, x)$.
(7) For an ideal $\mathfrak{a} \subseteq A$, define

$$\mathfrak{a}[[x]] = \left\{ \sum_{i=0}^{\infty} a_i x^i : a_i \in \mathfrak{a} \text{ for all } i \geqslant 0 \right\}.$$

Find an example of a ring A and an ideal $\mathfrak{a}$ of A such that the extension of $\mathfrak{a}$ in $A[[x]]$ is strictly contained in $\mathfrak{a}[[x]]$.